Mehmet Selman YILMAZ

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Computer Science undergraduate focused on AI and Natural Language Processing, with hands-on experience in machine learning and language technologies. I plan to pursue a Master's degree and, in the long term, a Ph.D. to deepen my expertise and research impact.

EDUCATION

Expected Jun 2026 **B.Sc. in Computer Science & Engineering (Minor in Finance)**, *Sabanci University*, **GPA:** 3.70 / 4.00 **Relevant Courses:** Machine Learning, Natural Language Processing, AI, Network Sceince, Deep Learning, Large Language Models

WORK EXPERIENCE

Aug – Sep 2025 **Artificial Intelligence Software Engineering Intern**, *Baykar*, Istanbul, Turkey

- Built a personalized recommendation system on Instagram, Twitter, and Next Sosyal datasets, using both real and synthetic user–content interactions (likes, comments, follows).
- Implemented a Two-Tower model for retrieval and a Wide & Deep model for re-ranking, leveraging embeddings extracted with TurkishBERTweet and metadata to improve quality.

Jul – Aug 2025 **Software Engineering Intern**, *BMC*, Izmir, Turkey

- Developed an AI-powered chatbot assistant using NestJS, Next.js, the OpenAI API, and SAP/ABAP integration for real-time inventory access and automated alerts.

Jun – Aug 2024 **Software Engineering Intern**, *Intertech*, Istanbul, Turkey

- Built a generative AI chatbot for financial insights using RAG architecture, integrating SQL, Next.js (frontend), and Node.js (backend).
- Automated the retrieval and processing of daily financial bulletins via Python web scraping, storing them in Azure Blob Storage for model input.

Jun 2023 – Jun 2024 **Assistant for Programming Fundamentals**, *Sabanci University*, Istanbul, Turkey

- Graded assignments, helping with core C++ concepts.

PROJECTS

Apr 2026 – Present **AutoBEME – Bounty-Driven Evolutionary Market Ensemble for NLP**

- Designed a novel multi-agent ensemble framework for low-resource NLP, combining bootstrap-sampled Classifier Chains, asymmetric penalty weighting, and market-based capital updates in a scikit-learn compatible pipeline.
- Outperformed fine-tuned frozen-transfer BERT baselines on Turkish text classification, improving Macro-F1 from 0.624 to 0.640.

Jan 2026 – Present **Dynamic Signed Network Analysis of the Turkish Parliament**

- Built a computational social science pipeline over 1.33M+ TBMM speeches (2002–2026), generating directed signed party-to-party networks, concept networks, and monthly political-economy panels across legislative terms 22–28.
- Developed validation and model-comparison workflows with a 733-row human-annotated gold set; AutoBEME-supported sentiment modeling achieved Macro-F1=0.640.

Jan 2026 – Present **BlueSky Turkish Politics & Science Feed Generator**

- Developed a real-time Turkish politics and science feed generation system using the AT Protocol firehose, BERTurk-based NLP processing.
- Designed a 5-stage NLP workflow including Turkish language detection, 768-dimensional embedding generation, zero-shot domain classification.

Jan – Mar 2026 **Transport 245 – New-Generation Logistics Network**

- Built a cross-platform logistics platform for drivers, institutions, and end users with location-based provider discovery and type filtering (Next.js frontend, Nest.js backend).
- Built modular REST APIs for onboarding, geospatial queries, and subscription-ready payments; improved performance with loading states and pagination for large provider lists.

Sep 2025 – Jan 2026 **EchoSpaceAR – AR-Based Audio Awareness**

- Developed real-time cue/HUD flow via AR UI integration in Unity and C# scripts; aligned design decisions with HCI principles.
- Performed server-side processing for environmental sound classification with YAMNet and speech-to-text extraction with Whisper from stereo audio streams.

Sep 2025 – Jan 2026 **Music Emotion Recognition on DEAM – Segment CNN & Attention Pooling**

- Built a pipeline for valence/arousal regression on DEAM with log-mel spectrograms, 5s segmentation (50% overlap), and train-only global z-normalization.
- Improved song-level performance via attention pooling and an attention-head ensemble; best model achieved test MSE=0.0114, Pearson=0.721, and CCC=0.689.

Sep 2024 – Jan 2025 **Intent Detection & Slot Filling in NLP**

- Reproduced and evaluated Slot-Gated BiLSTM on the SNIPS dataset, achieving 97% accuracy for intent detection and 79% for slot filling.

Sep 2024 – Jan 2025 **Instagram Influencer Classification & Like Prediction**

- Built a stacking classifier (Random Forest + Logistic Regression) achieving 86.7% F1-score; applied SMOTE and domain-specific preprocessing on noisy social media data.
- Predicted likes using a RF Regressor for skewed distributions with feature engineering.

SKILLS

Languages Turkish (Native), English (Advanced)
Programming Python, C++, SQL, R, React.js, Node.js, Docker, Git, Azure
AI & Data TensorFlow, PyTorch, Scikit-learn, OpenAI API, RAG, NLTK, SVD, Pandas, NumPy

ACTIVITIES

Jun 2023 – Apr 2025 **Vice President**, *IEEE Computer Society, Sabanci University*

- Co-led the organization of CompTalks, an event series that hosted industry professionals.

VOLUNTEER WORK

Feb – Jun 2022 **Sabanci University Civic Involvement Projects**, Istanbul, Turkey

- Volunteered in weekly educational activities with elementary school students as part of Sabanci University's civic involvement program.

INTERESTS

Table tennis, cinema, travel, chess, and billiards